

TrES-5b

R-0935

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_240803 · created 2026-07-29T12:01:45+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460525.8823 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.031
Transit depth (Rp/Rs)^2	2.33 +/- 0.95 [%]
Orbital Inclination (inc)	84.82 +/- 2.19
Airmass coefficient 1 (a1)	0.87 +/- 0.02
Airmass coefficient 2 (a2)	0.116 +/- 0.035
Scatter in the residuals of the lightcurve fit is	2.32 %
Best Comparison Star	#3 - [224, 170]
Optimal Method	PSF photometry
Transit Duration (day)	0.074 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.153 ± 0.031 vs 0.142 (deviation 0.35σ)
Duration fit vs published	0.074 d vs 0.0758 d (deviation 0.14σ)
Mid-time O–C	7.9 min
Depth SNR	4.9
Residual scatter	2.32 %

Light curve

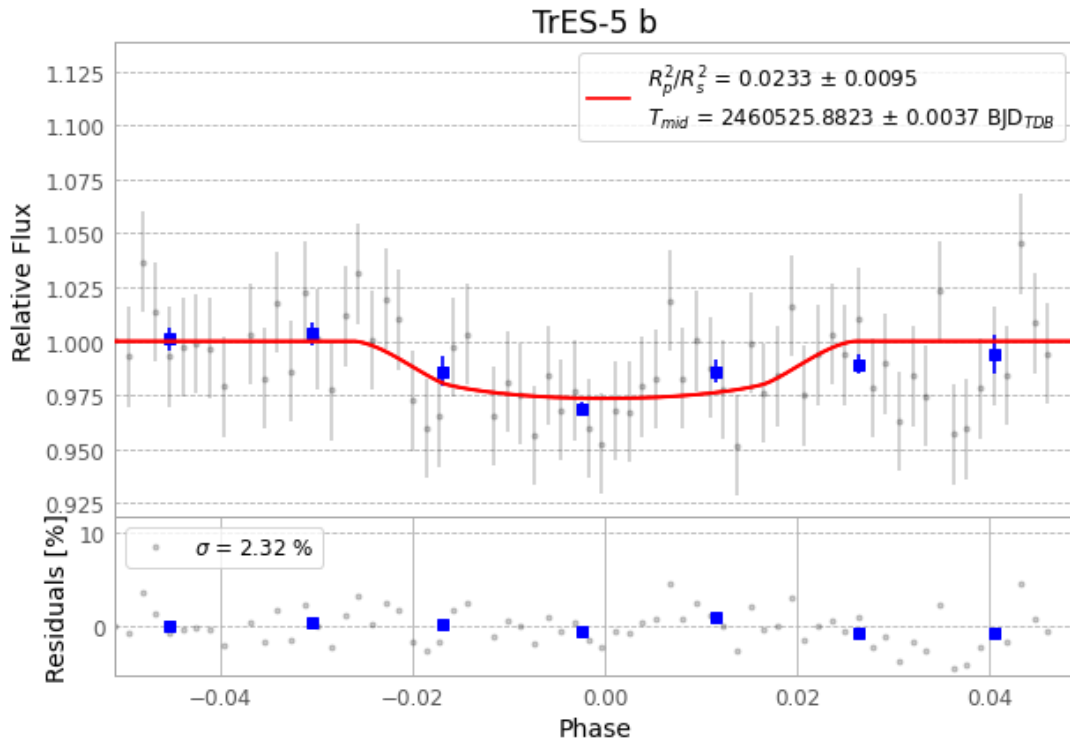


Figure 1 — FinalLightCurve_TrES-5 b_2024-08-03.png (SHA-256 74b83161dc194b3e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [368, 267], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e9bd7e7c67c91460...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

72 FITS frames (47 MB), TRES-5240803071815.FITS → TRES-5240803105115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-240803.log (747ede3d423c...), FinalLightCurve_TrES-5 b_2024-08-03.png (74b83161dc19...), FinalLightCurve_TrES-5 b_2024-08-03.csv (09c6a8fa0cf7...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 12:01Z.